

Multiple Choice (16 marks)

1. Which types of functions grow the slowest?
 - (a) $O(N)$
 - (b) $O(\log \log N)$
 - (c) $O(N!)$
 - (d) $O(\log N)$
 - (e) $O(N^N)$
2. If your training loss increases with number of epochs, which of the following could be a possible issue with the learning process?
 - (a) Regularization is too high and model is underfitting
 - (b) The number of epochs is too high
 - (c) Step size is too small
 - (d) Step size is too large
 - (e) The number of epochs is too low
3. Which among the following prevents overfitting when we perform bagging?
 - (a) The use of all data without sampling
 - (b) The practice of validation performed on a random subset of classifiers trained
 - (c) The use of underfitting to counterbalance overfitting
 - (d) The use of weak classifiers
 - (e) The use of classification algorithms which are prone to overfitting
4. What is the number of labelled forests on 8 vertices with 5 connected components, such that vertices 1, 2, 3, 4, 5 all belong to different connected components?
 - (a) 620
 - (b) 520
 - (c) 220
 - (d) 120
 - (e) 820
5. Any set of Boolean operators that is sufficient to represent all Boolean expressions is said to be complete. Which of the following is NOT complete?
 - (a) {NAND}

- (b) {NOT, NAND}
 - (c) {AND, OR}
 - (d) {NOT, OR}
 - (e) {XOR, NOT}
6. Adding more basis functions in a linear model, pick the most probably option:
- (a) Decreases model bias
 - (b) Increases estimation bias
 - (c) Doesn't affect bias and variance
 - (d) Increases both bias and variance
 - (e) Decreases variance
7. Statement 1| The softmax function is commonly used in mutliclass logistic regression.
Statement 2| The temperature of a nonuniform softmax distribution...
- (a) False, False
 - (b) True, Statement 2 is not related to Statement 1
 - (c) False, Both statements are related to each other
 - (d) Statement 2 is true, but Statement 1 is irrelevant
 - (e) Statement 1 is true, but Statement 2 is irrelevant
8. Which of the following is not a security exploit?
- (a) Malware
 - (b) Keylogging
 - (c) Data breach
 - (d) Authentication
 - (e) Eavesdropping

True / False (14 marks)

Answer True or False and justify briefly.

1. True or False: The correct answer to 'In Python 3, what is ['a', 'Chemistry', 0, 1][1]?' is 'Chemistry'.
2. True or False: The correct answer to 'In a Gigabit Ethernet LAN, the average size of a frame is 1000 bytes. If a noise of 2ms occurs on the LAN, how many frames are destroyed?' is '100'.

3. True or False: The correct answer to 'Statement 1| Highway networks were introduced after ResNets and eschew max pooling in favor of convolutions. Statement 2| DenseNets...' is 'True, False'.
4. True or False: The correct answer to 'What is the number of labelled forests on 10 vertices with 5 connected components, such that vertices 1, 2, 3, 4, 5 all belong to...' is '40000'.
5. True or False: The correct answer to 'Encapsulating Security Payload (ESP) belongs to which Internet Security Protocol?' is 'Internet Key Exchange Protocol'.
6. True or False: The correct answer to 'Suppose a convex polygon has 26 faces and 39 edges. How many vertices does it have?' is '15'.
7. True or False: The correct answer to 'Statement 1| Industrial-scale neural networks are normally trained on CPUs, not GPUs. Statement 2| The ResNet-50 model has over 1 billion...' is 'False, True for ResNet-101 model only'.

Long Form Response (10 marks)

Provide thorough reasoning and reference key steps.

1. Write a function to find the sum of geometric progression series.

End of Paper